IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- Problems you want to find answers

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - The data was collected using SpaceX REST API and web scraping of SpaceX launch data from Wikipedia.
- Perform data wrangling
  - Data was cleaned, merged, and processed by handling missing values, encoding categorical variables and preparing features for analysis.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Classification models were built, tuned using cross-validation and evaluated to compare model performance.

Data Collection

Get Data → JSON to Data Frame → Data Extraction → Saved to CSV

Get Data	JSON to Data Frame	Data Extraction	Saved to CSV
• Python request library was used to obtain past launch data via REST API's.	• JSON object was normalized and converted to tabular data in a data frame.	• Functions were built and used to extract launch sites, cores and payloads.	• Data was stored as a csv file.

Data Collection – SpaceX API

- The notebook can be found on Github here.

Sent HTTP GET Request to SpaceX REST API

Verified successful responses via status codes

Parsed JSON responses for processing.

Data Collection - Scraping

- The notebook can be found on Github here.

Request Web Page

↓

Parse HTML

↓

Extract Launch Table

↓

Clean & Normalize Data

↓

Save to CSV

Data Wrangling

- Missing values were identified and handled
- Irrelevant columns were removed
- Categorical variables were encoded
- Cleaned Data was prepared for analysis
- The notebook can be found on Github here.

Load Raw Data

Handle Missing Values

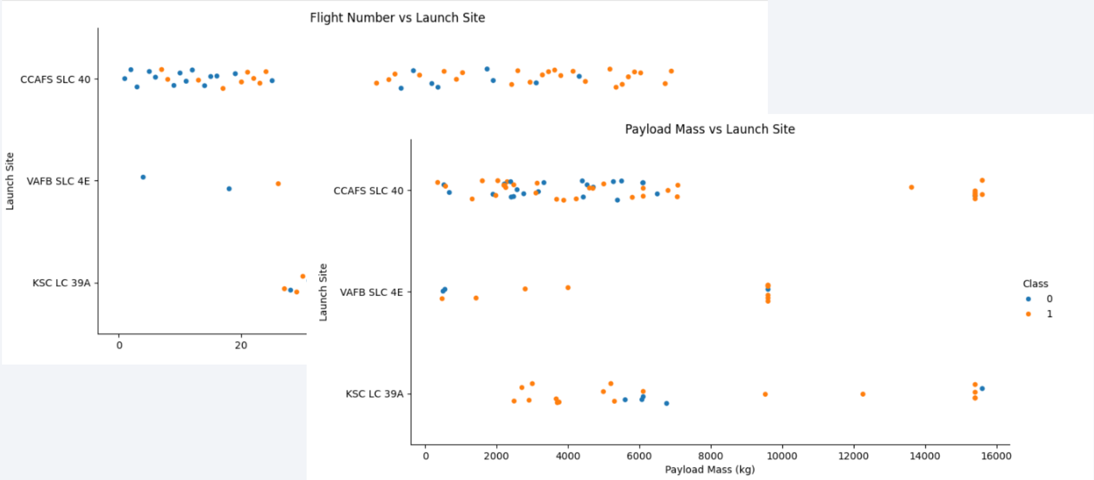
Filter and Select Columns

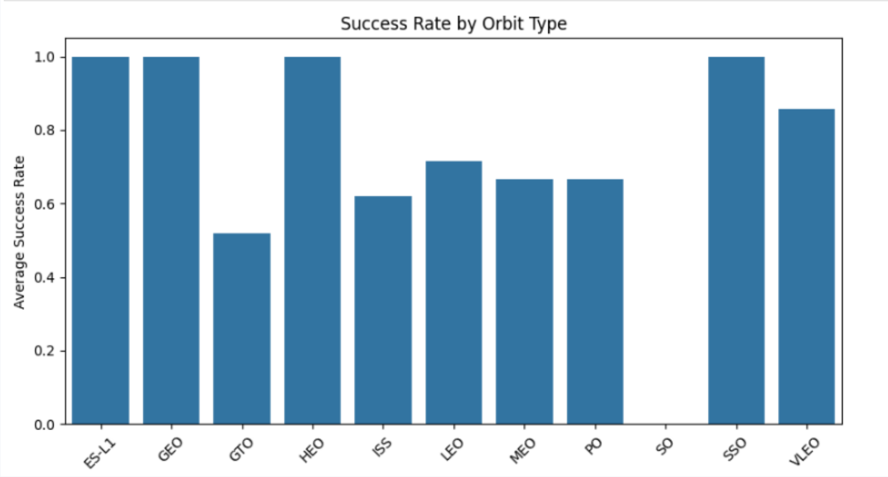
Encode categorical Data

Clean Dataset Saved

EDA with Data Visualization

- Scatter plots were used to analyze relationships between flight number, payload mass, orbit type and launch site versus launch success.
- Bar charts were used to compare average success rates across orbit types
- Line charts were used to observe yearly trends in launch site success rates.
- These charts were chosen to quickly identify patterns, correlations, and trends related to mission success.
- See: [GitHub]()





EDA with SQL

- Queried launch records to analyze launch success rates by site, orbit, and booster type.
- Used GROUP BY and aggregate functions to calculate counts and average success rates.
- Filtered data using WHERE clauses to isolate specific launch sites and mission outcomes.
- Ordered and summarized results to identify patterns and trends in launch performance.
- GitHub here.

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

- DID not complete. Lab wouldn’t run.

Predictive Analysis (Classification)

- Built multiple classification models:
    - Logistic Regression
    - SVM
    - Decision Tree
    - KNN
- Split Data into training and testing sets
- Tuned hyperparameters using GridSearchCV with cross-validation
- Evaluated models using accuracy and confusion matrix
- Selected best performing model based on validation accuracy
- Predicitve Analysis can be found on GithHub here.

Results

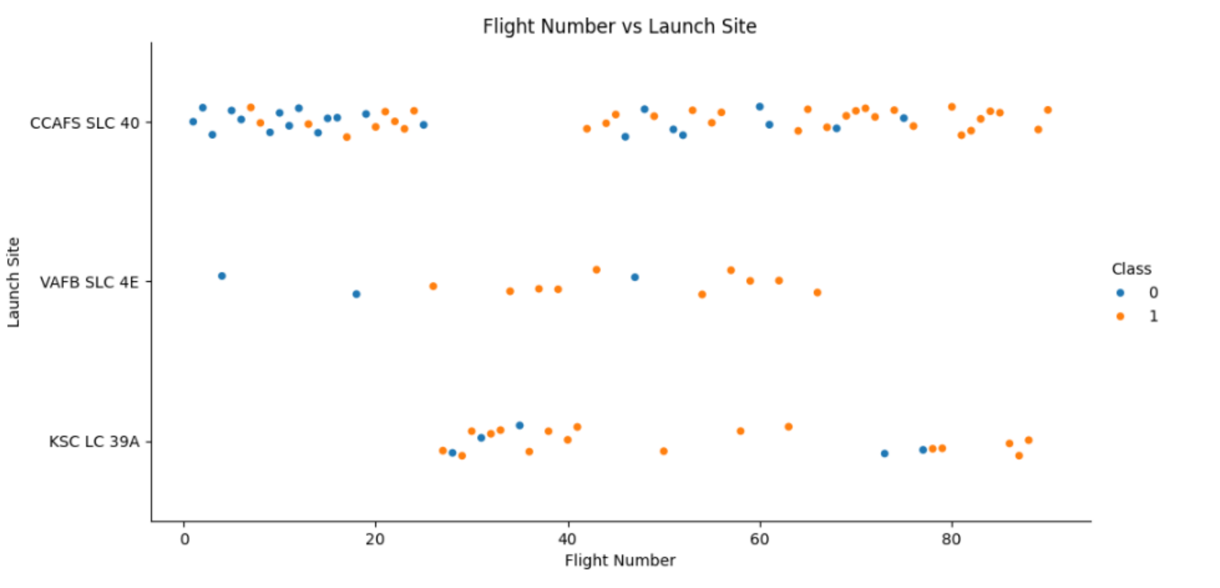
- Exploratory data analysis results.
  - Launch success rates varies by launch site
  - Higher payload ranges show improved success rates
  - Certain booster versions outperform others
- Interactive analytics demo in screenshots
  - Not completed
- Predictive analysis results
  - Multiple classifiers were evaluated
  - Tuned models improved prediction accuracy
  - Best model achieved highest validation accuracy using cross-validation

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- The scatter plot shows launches by flight number across different launch sites, colored by mission and outcome.
- Early flight numbers contain more failed launches.
- As flight numbers increase, successful launches become dominant across all sites.
- Indicated that launch success improves over time, regardless of launch site.



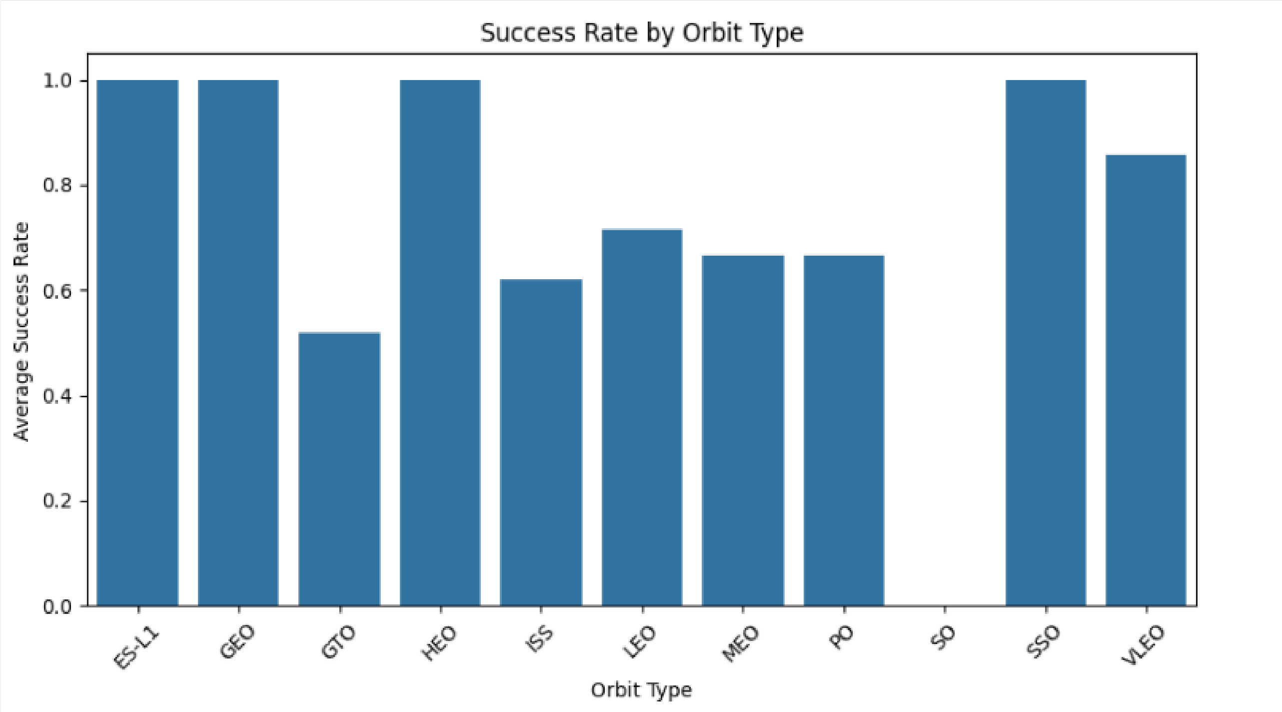
Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

- Show the screenshot of the scatter plot with explanations

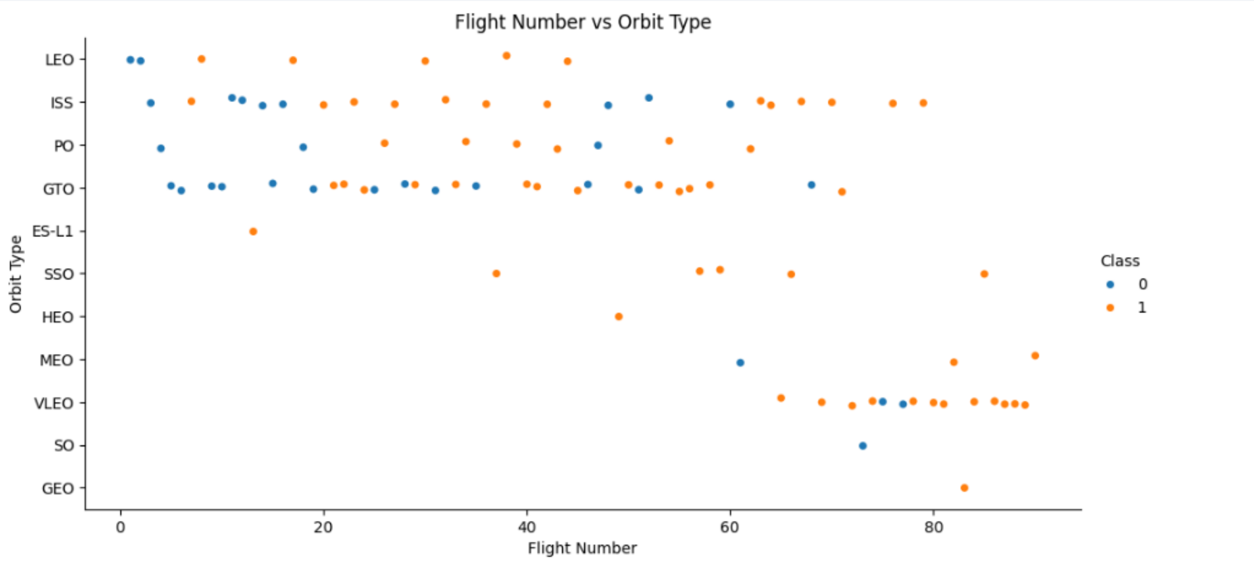
Success Rate vs. Orbit Type

- A bar chart was used to compare average launch success rates across difference orbit types.
- Orbits such as LEO and ISS show higher success rates compared to others.
- This suggests that mission success is strongly influenced by orbit type, with more frequently flown orbits achieving higher reliability.



Flight Number vs. Orbit Type

- The scatter plot shows launch outcomes across orbit types over increasing flight numbers.
- **LEO and ISS orbits** show a clear increase in successful launches as flight number increases.
- Early missions exhibit more failures, while **later flights are predominantly successful**, indicating improved reliability.
- **GTO missions** show less consistency, suggesting higher mission complexity and risk.

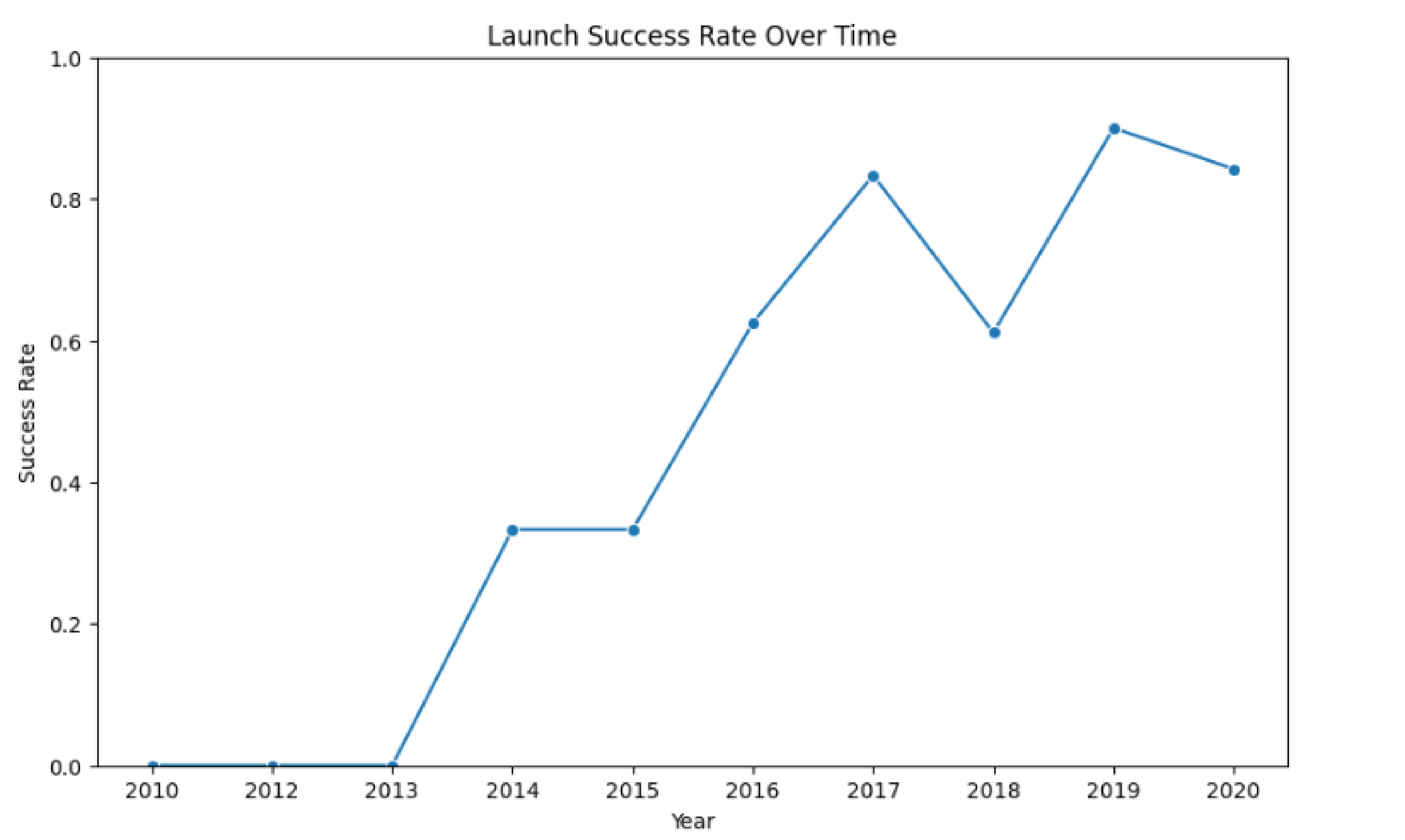


Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

- Show the screenshot of the scatter plot with explanations

Launch Success Yearly Trend



All Launch Site Names

```
%%sql
SELECT DISTINCT "Launch_Site"
FROM SPACEXTBL
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%%sql
SELECT *
FROM SPACEXTBL
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
%%sql
SELECT SUM("Payload_Mass__kg_") AS total_payload_mass
FROM SPACEXTBL
WHERE "Customer" = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

total_payload_mass
45596

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG("Payload_Mass__kg_") AS avg_payload_mass
FROM SPACEXTBL
WHERE "Booster_Version" = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

avg_payload_mass
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

```
%%sql
SELECT MIN("Date") AS first_success_ground_pad
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

first_success_ground_pad
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
SELECT DISTINCT "Booster_Version"
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS__KG_" > 4000
  AND "PAYLOAD_MASS__KG_" < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
%%sql
SELECT "Mission_Outcome", COUNT(*) AS count
FROM SPACEXTBL
GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%%sql
SELECT DISTINCT "Booster_Version"
FROM SPACEXTBL
WHERE "Payload_Mass__kg_" = (
    SELECT MAX("Payload_Mass__kg_") FROM SPACEXTBL
);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
%%sql
SELECT
  substr("Date", 6, 2) AS month,
  "Landing_Outcome",
  "Booster_Version",
  "Launch_Site"
FROM SPACEXTBL
WHERE substr("Date", 1, 4) = '2015'
  AND "Landing_Outcome" IN ('Failure (drone ship)', 'Precluded (drone ship)')
ORDER BY "Date";
```

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
06	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS count
FROM SPACEXTBL
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY count DESC;
```

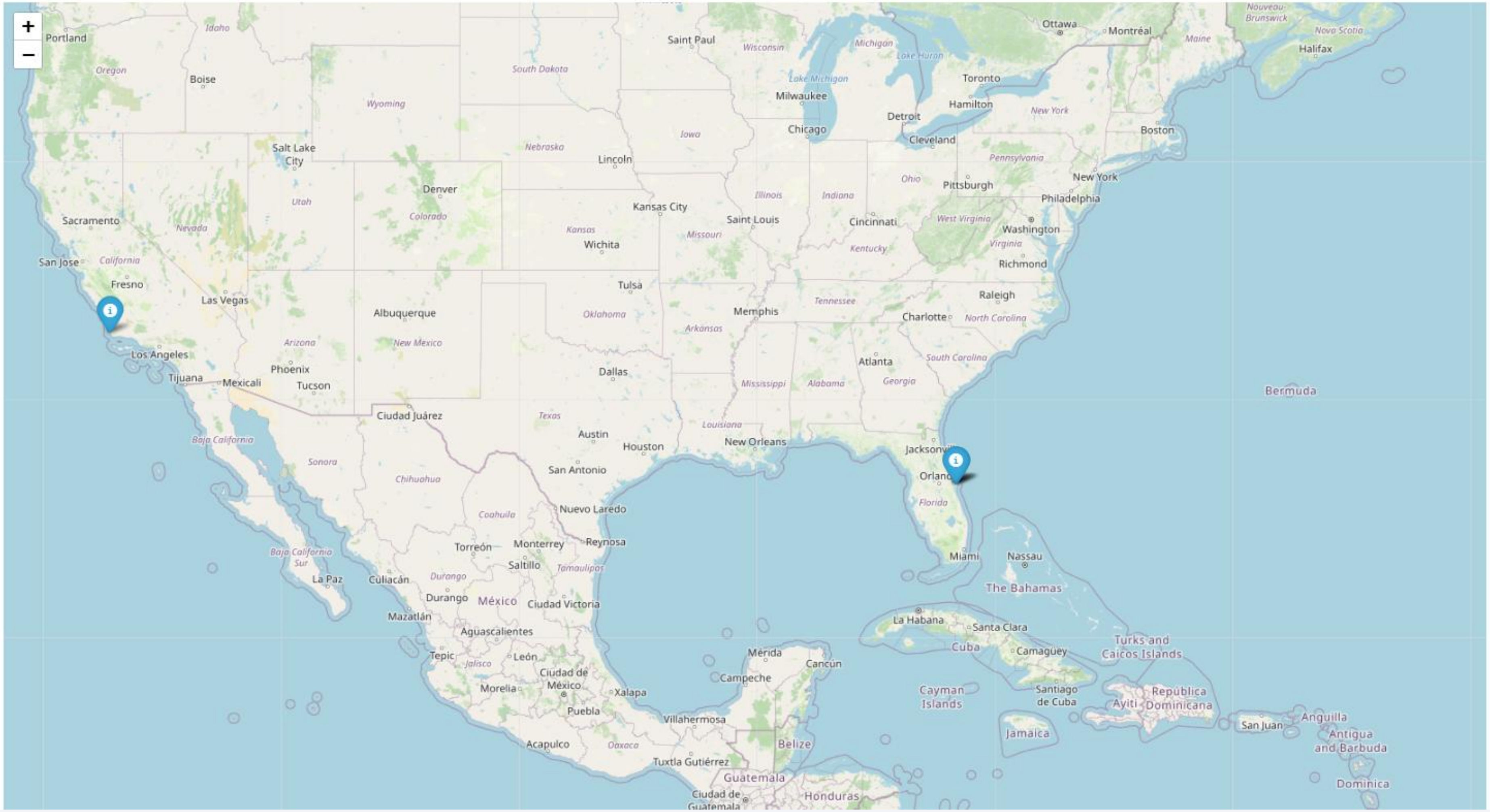
```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

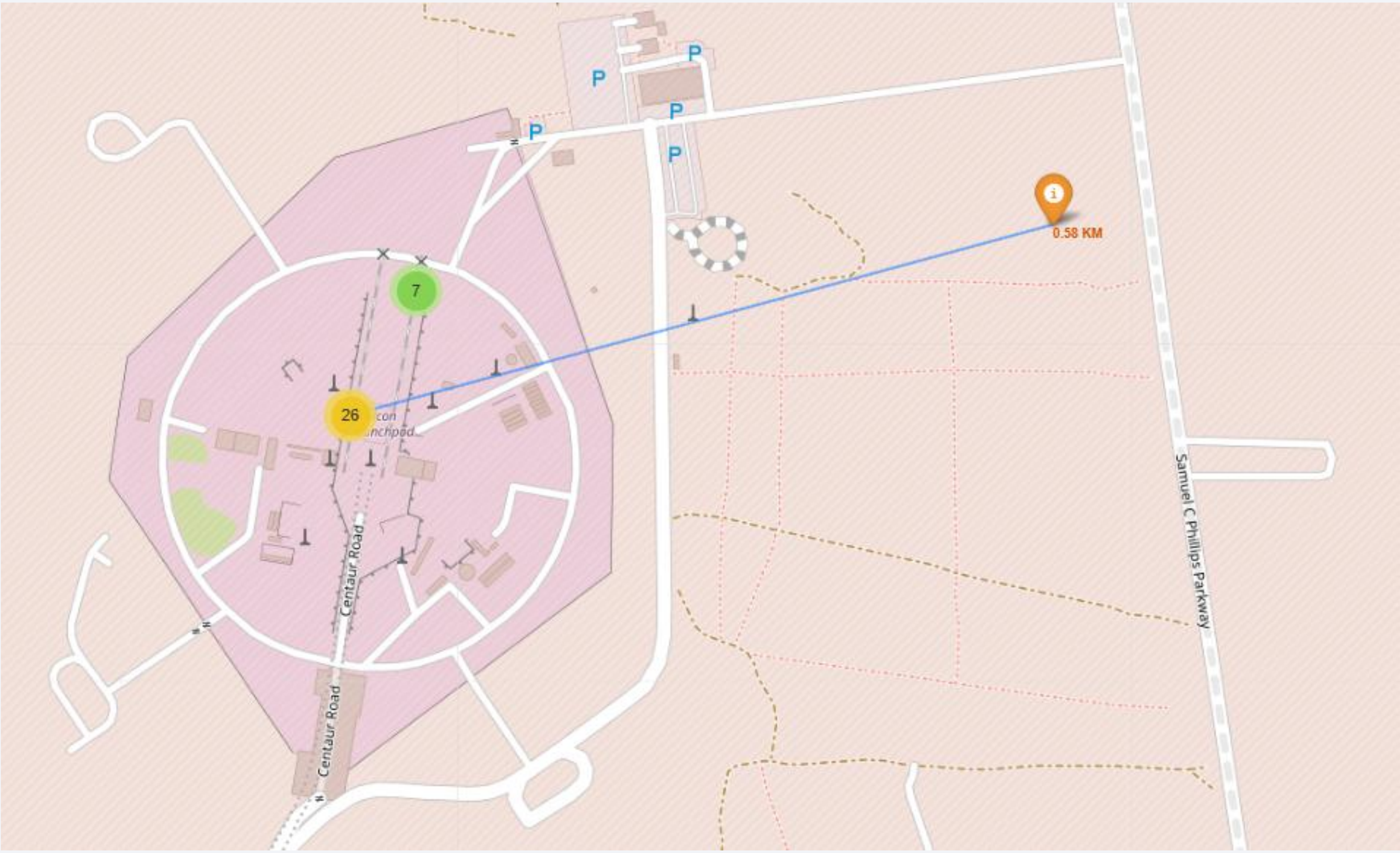
Launch Sites Proximities Analysis

All Launch Sites



Success and Failed launches for each site

Distance between Launch Site to Proximity



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

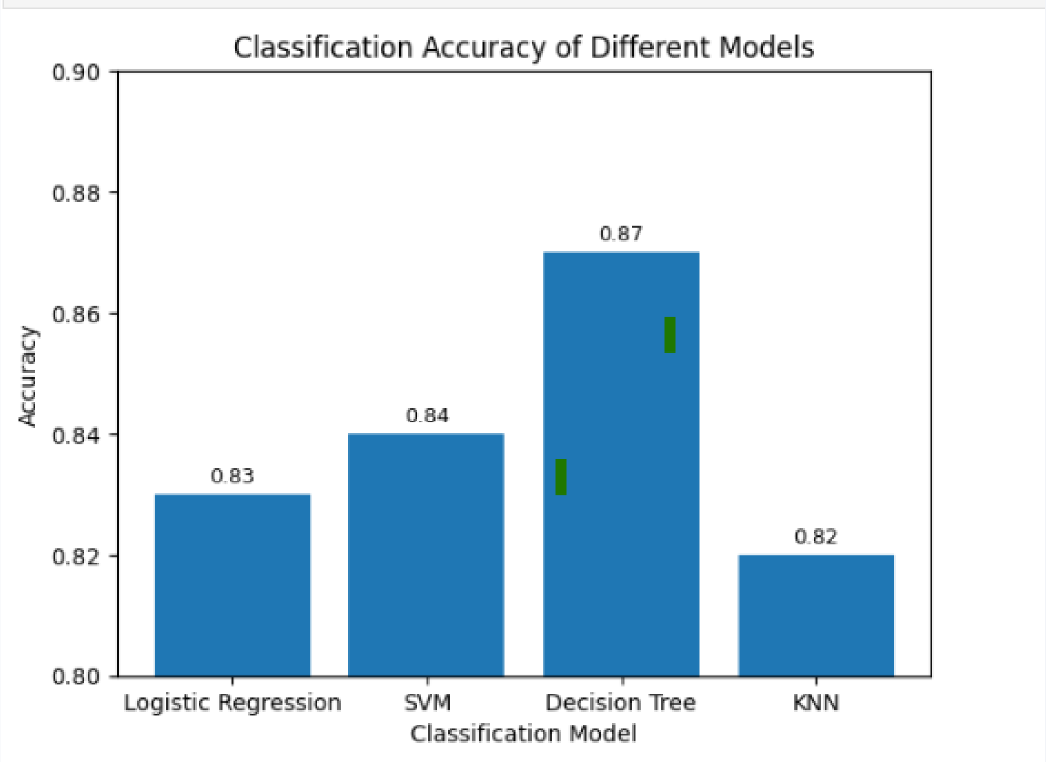
- Did not complete. Lab wouldn’t run.

Section 5

Predictive Analysis (Classification)

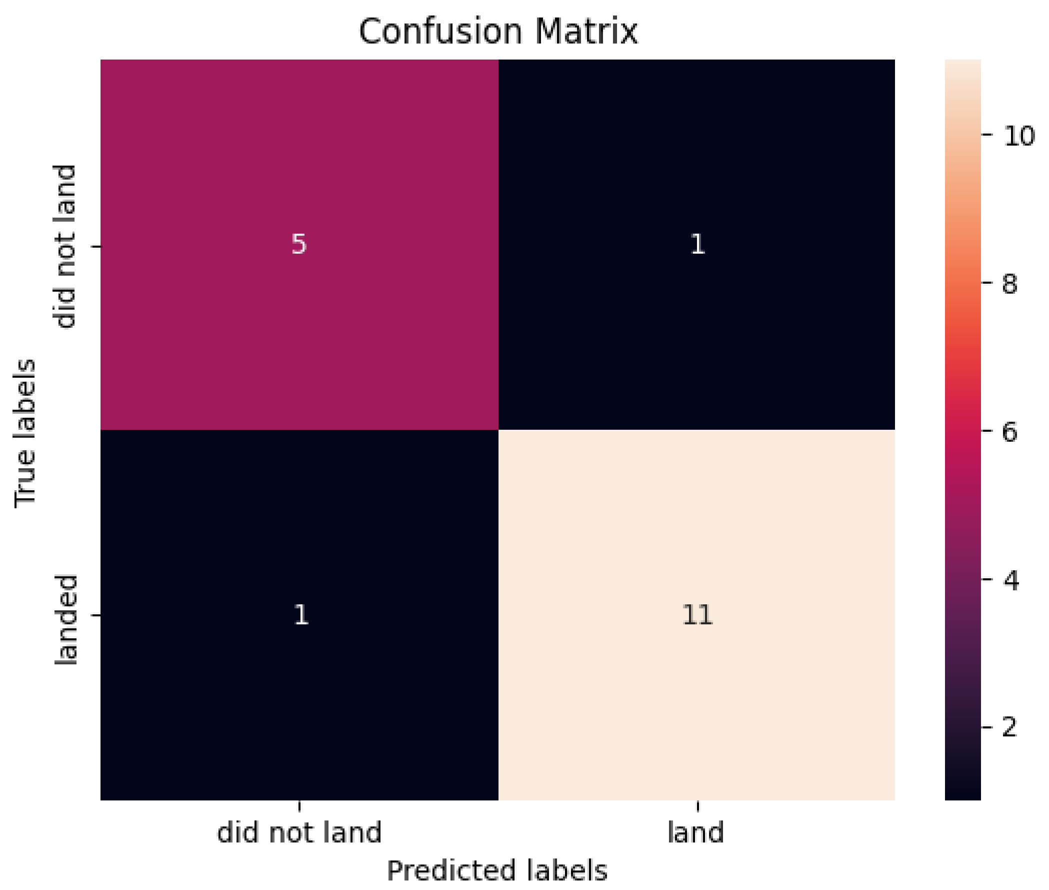
Classification Accuracy

As the chart shows, decision tree showed the highest accuracy for classification.



Confusion Matrix – Decision Tree

- **True Negative (5):** The model correctly predicted “did not land” when the rocket did not land.
- **False Positives (1):** The model predicted “landed” when the rocket did not land.
- **False Negative(1):** The model predicted “did not land” when the rocket did land.
- **True Positive (11):** The model predicted landed when the rocket landed.



Conclusions

- Multiple classification models were evaluated to predict successful rocket landings.
- Cross-validation results showed that Decision Tree model achieved the highest classification accuracy.
- Visualization of model accuracies confirmed that all models performed similary, with the Decision Tree performing slightly better.
- The confusion matrix for the best model demonstrated a high number of correct predictions with only a small number of missclassifications.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!